

| Education           |                        |                                                |         |
|---------------------|------------------------|------------------------------------------------|---------|
| Year                | Degree/Certificate     | Institute                                      | CPI/%   |
| Oct 2022 – Aug 2026 | B.Tech in CS & IT      | Ajay Kumar Garg Engineering College, Delhi NCR | 7.25/10 |
| 2021 – 2022         | Senior Secondary (XII) | New Era School, Delhi NCR                      | 91.2%   |
| 2017 – 2020         | High School (X)        | SJN International School, Coimbatore           | 92.6%   |

Experience

Google Summer of Code 2025 Contributor | *P4 Language Consortium, Stanford, CA* (May 2025 – Sept 2025)

SpliDT Project: Scaling Stateful Decision Tree Algorithms in P4

*P4, Python, P4Runtime, Docker, C++, Intel Tofino*

- Switch-Native Architecture:** Architected a framework to compile ML decision trees directly into P4 data planes, implementing Partitioned Decision Trees (PDTs) with TCAM-efficient Subtree IDs (SIDs) to resolve hardware ASIC constraints.
- Validation & Optimization:** Engineered a Jinja2-based P4Runtime testing pipeline (Tofino/Mininet) improving deployment reliability by 30%, and optimized SID recirculation to achieve 10 Gbps line-rate throughput with 20% lower inference latency.

Open Source Contributions

- Project-HAMi/HAMi:**
  - #2292 Corrected hardware allocation logic for Kunlun and AWS Neuron AI accelerators to prevent invalid state generation
  - #2307 Eliminated node-level deadlocks by strictly enforcing lock releases during failed device-binding operations
  - #2259 Stabilized cluster scheduling by adding safe state handling for deleted ResourceQuota tombstones
  - #2204 Prevented metrics panics by guarding hardware memory ratio calculations against unknown total memory states
  - #2221 fix(workflow): skip non-translatable issue comments
- WasmEdge/WasmEdge:**
  - #4470 fix(wasi): resolve 100% CPU usage on macOS due to clock mismatch
  - #4440 Upgrade Alpine base to 3.23 and restore static LLD support
  - #4466 feat(docker): add standalone alpine-base image build pipeline
  - #4452 docs(README): fix broken introduction and blog links
- facebook/bpfilter:**
  - #507 build: update clang-tidy brace config, fix opts.c/set.c, and NOLINT dump.c

Key Projects

**Cutable** | *Go, Next.js, PostgreSQL, OpenRouter, E2B, WebSockets, Nomad* [\(Project Link\)](#)

**Objective:** Generate editable React applications from text and visual requirements.

**Approach:** Orchestrated OpenRouter tools and E2B sandboxes in Go; streamed code and live previews to Next.js over WebSockets.

**Impact:** Ships multimodal BYOK generation, persistent projects, and health-gated deployments.

**DMIE** | *Next.js, TypeScript, Python, FastAPI, Vector DB, LLM* [\(Project Link\)](#)

**Objective:** Determine whether a startup idea already exists by comparing it against more than **500,000** company records.

**Approach:** Built a scheduled scraper that keeps the startup dataset current; combined LLM query with vector search and backend filters.

**Impact:** Raised USD **10,000** and serves more than **100** users worldwide.

**XNIC v1** | *C, Linux Kernel, PCI, DMA, NAPI, QEMU, DPDK (rte\_ethdev)* [\(Project Link\)](#)

**Objective:** Build and qualify a defensible clean-room Linux Ethernet driver with observable packet paths, ownership invariants, fault recovery, and a gated user-space DPDK forwarding path.

**Approach:** Implemented PCI probe/remove, BAR0 MMIO, coherent and streaming DMA rings, interrupt-driven NAPI, TX queue backpressure, ethtool counters, and serialized watchdog/reset recovery in C; added staged fault injection, KFENCE and sparse qualification, reproducible QEMU/HVF automation, Wireshark-compatible captures, a W5500 SPI driver path, and an rte\_ethdev L2 forwarder.

**Impact:** Qualified **10,100** RX and TX ring wraps across **646,400** packets at zero loss, **1,000** interface cycles, **100** driver rebinds, and **100** reset-under-traffic cycles; validated DPDK partial-TX cleanup and signal-safe shutdown with the net\_pcap virtual PMD.

**Do-It** | *Go, SSE, HTTP, Raspberry Pi, Local-first* [\(Project Link\)](#)

**Objective:** Provide private, synchronized notes and media across devices on a local network.

**Approach:** Packaged a lightweight Go SSE and HTTP server for Raspberry Pi, routers, and repurposed phones.

**Impact:** Delivers self-hosted, account-free synchronization without sending household data to a cloud service.

**eBPF-mcp-tracer** | *Python, bpftrace, eBPF, Linux, MCP* [\(Project Link\)](#)

**Objective:** Let AI debugging agents inspect Linux performance without granting unrestricted eBPF execution.

**Approach:** Implemented a strict probe allowlist, dry-run validation, and execution timeouts around bpftrace; exposed the guarded tracing workflow through MCP.

**Impact:** Enables kernel-level diagnostics while reducing command-injection and runaway-probe risk.

**Los-Tecnicos** | *Go, TypeScript, Rust, Soroban, Raspberry Pi, ESP32* [\(Project Link\)](#)

**Objective:** Build a decentralized energy grid that tokenizes renewable-energy production on the Stellar ecosystem.

**Approach:** Built a Go backend and **4** Soroban smart contracts; integrated a Raspberry Pi/ESP32 hardware mesh with the platform.

**Impact:** Won Asia-Pacific Web3 bounties and received recognition from the Government of India.

**Open Claw Lab** | *Node.js, OpenClaw, Gemini, Oracle Cloud, systemd, Telegram* [\(Project Link\)](#)

**Objective:** Discover high-signal engineering opportunities while preventing unsupported or stale leads from reaching the user.

**Approach:** Built a multi-agent discovery and independent-review pipeline around typed lead records; added deterministic freshness, eligibility, contact relevance, relocation evidence, and deduplication gates.

**Impact:** Delivers an evidence-backed Telegram digest while rejecting guessed contacts, unsupported geography, stale roles, and duplicate leads.

**Slix** | *React, Go, Azure OpenAI, REST API* [\(Project Link\)](#)

**Objective:** Combine movie streaming, reviews, and machine-generated sentiment summaries in one full-stack application.

**Approach:** Built the frontend in React and the service layer in Go; added Azure OpenAI sentiment analysis for curator reviews.

**Impact:** Gives viewers a concise representation of a movie's tone before watching.

**p4Lens** | *React, FastAPI, Docker, React Flow, P4* [\(Project Link\)](#)

**Objective:** Make P4 programs easier to learn and debug through an interactive visualization of their processing pipeline.

**Approach:** Built a parser for P4-14 and P4-16 programs; rendered headers, parsers, controls, and deparsers as interactive React Flow topologies.

**Impact:** Renders pipeline diagrams in under **200** ms and reduces control-plane debugging effort by approximately **40** percent.

**Moodish** | *JavaScript, Node.js, OpenAI API, MCP* [\(Project Link\)](#)

**Objective:** Rank meal options from a user's craving, budget, and dietary constraints while keeping the recommendation explainable.

**Approach:** Built intent extraction and deterministic ranking around an AI-generated summary; added a recommendation trace and confirmation-gated cart preview.

**Impact:** Produces safer, inspectable food recommendations instead of opaque model output.

**Myprod** | *Go, Nomad, Traefik, WireGuard, Docker, Netlify DNS, SOPS* [\(Project Link\)](#)

**Objective:** Turn manually provisioned VPS machines into a low-overhead personal compute pool for backend services.

**Approach:** Built a Go CLI and operator dashboard around Nomad scheduling, Traefik ingress, and WireGuard networking; added guarded node lifecycle controls, application registration, live resource metrics, and idempotent Netlify DNS management.

**Impact:** Provides one control surface for deploying applications and managing heterogeneous free-tier and temporary compute capacity.

## Technical Skills

- **Language:** Python, C, C++, JavaScript, TypeScript, Go, P4
- **Libraries/Frameworks:** NodeJS, Numpy, Matplotlib, eBPF, gRPC, Protobufs, Next.js, Scapy, MLIR, LLVM, Sklearn, React, Web Services
- **Tools/Platforms:** Docker, CI/CD, VS Code, UTM, GitHub, GitLab, DaVinci Resolve, Linux (x86\_64, ARM64 & AMD64)

- **Databases:** SQL, MongoDB, GraphQL, Redis, PostgreSQL, MySQL

Positions of Leadership

- **Technical Writer, Cloud Native Community Group, New Delhi** (May 2024 – Dec 2024)
  - Led a team of **6+** content writers and **4** designers to deliver documentation and event management support.
  - Partnered with marketing to revamp social media presence across LinkedIn, Instagram, and X, achieving a **60%** bump in engagement.
- **Co-Organizer & Anchor, Google Developer Groups & GDSC WOW 2024, Delhi NCR** (Nov 2023 – Nov 2025)
  - Co-led **10+** tech events & anchored the GDSC WOW flagship event at IIIT Delhi, with the **31+** crew members.
  - Engaged a live audience of **800+** at WOW and **500+** developers overall, boosting participation by **30%** and visibility by **40%**.

Publications

- R. Bharadwaj, **S. Jha**, V. Malyan, R. Gupta, “Trusture: A Blockchain-Based Decentralized Framework for Secure, Transparent, and Auditable NGO Transactions,” *2026 International Conference on Computing, Sciences and Communications (ICCSC)*, Ghaziabad, India, Feb. **2026**.  
[IEEE Xplore](#)  [DOI: 10.1109/ICCSC67078.2026.11468597](#)

Certifications & Honors

|                 |                                                                                                                                                                                                                                                                    |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Certifications  | • <b>Open Science 101</b> – NASA    • <b>Data Science for Engineers</b> – NPTEL    • <b>Enhancing Soft Skills &amp; Personality</b> – NPTEL    • <b>IIRS Satellite Remote Sensing</b> – ISRO                                                                       |
| Honors & Awards | • <b>2x</b> AKTU State Men Singles TT Gold ( <b>2023, 2024</b> )    • International Robotics Olympiad Bronze, Thailand ( <b>2019</b> )    • Merit Certification at Carnic Model United Nation ( <b>2021</b> )    • NCC Cadet, National Cadet Corps ( <b>2019</b> ) |